Vectors

Chapter Overview

1. Distance between two points

How far apart are $(1, 0, 4)$ and $(-3, 5, 9)$?

2. Column vectors and unit vectors

$$\begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} = \mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$$

3. Magnitude and direction

Find the angles that $2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ makes with the positive coordinate axes.

4. Geometric problems

Extend the Year 1 vector methods to three dimensions.

5. Applications to mechanics

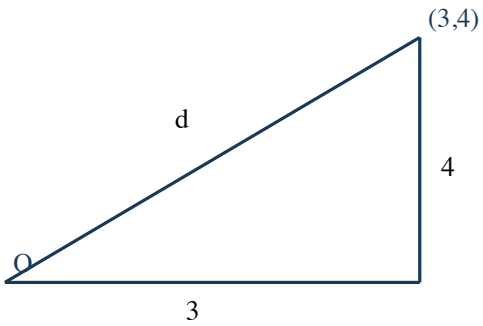
Use $\mathbf{F} = m\mathbf{a}$ with three-dimensional vectors. Relate displacement to distance and velocity to speed.

This chapter reviews three-dimensional coordinates and extends familiar vector results from 2D to 3D.

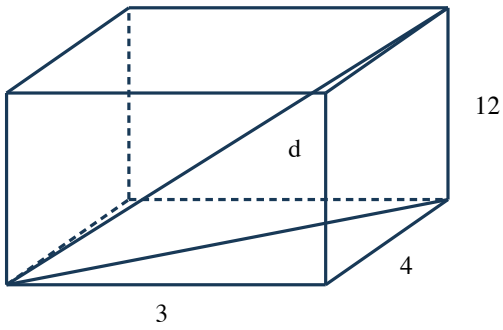
Distance from the origin and magnitude of a vector

Solution 3

In 2D, find the distance of $(3, 4)$ from the origin.



In 3D, find the distance of $(3, 4, 12)$ from the origin.



2D distance

$d = \sqrt{3^2 + 4^2} = 5$

3D distance

Base diagonal: $\sqrt{3^2 + 4^2} = 5$. Then use the highlighted triangle:

$d = \sqrt{5^2 + 12^2} = 13.$

Equivalently, in one step:

$d = \sqrt{3^2 + 4^2 + 12^2} = 13.$

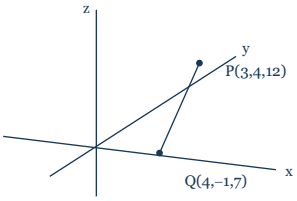
For $\mathbf{a} = (x, y, z)^T$, its magnitude is

$|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$

Distance between two 3D points

Solution 4

For $P(3, 4, 12)$ and $Q(4, -1, 7)$, find PQ.



PQ

$\overrightarrow{PQ} = (1, -5, -5)^T$

$PQ = \sqrt{1 + 25 + 25} = \sqrt{51}$

$d = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$.

Δx means change in x. Squaring means a positive or negative coordinate change gives the same result.

Quick checks

1. Distance from $(4, 0, -2)$ to the origin.

1 $\sqrt{4^2 + 0^2 + (-2)^2} = \sqrt{20}$

2. Magnitude of $(5, 4, -1)^T$.

2 $\sqrt{25 + 16 + 1} = \sqrt{42}$

3. Distance between $(0, 4, 3)$ and $(5, 2, 3)$.

3 $\sqrt{25 + 4} = \sqrt{29}$

4. Distance between $(1, 1, 1)$ and $(2, 1, 0)$.

4 $\sqrt{1 + 0 + 1} = \sqrt{2}$

5. Distance between $(-5, 2, 0)$ and $(-2, -3, -3)$.

5 $\sqrt{9 + 25 + 9} = \sqrt{43}$

Test Your Understanding So Far...

Solution 5

1. Find the distance from the origin to $P(7,7,7)$.

1

$$OP = \sqrt{7^2+7^2+7^2} = 7\sqrt{3}$$

2. $A = (5,3,-8)$ and $B = (1,k,-3)$. Given $AB = 3\sqrt{10}$, find the possible values of k.

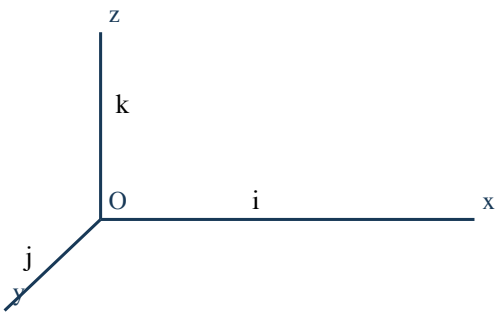
2

$$\sqrt{4^2+(k-3)^2+5^2} = 3\sqrt{10}$$
$$\sqrt{k^2-6k+50} = 3\sqrt{10}$$
$$k^2-6k+50 = 90$$
$$k^2-6k-40 = 0$$
$$(k+4)(k-10) = 0$$
$$k = -4 \text{ or } 10.$$

Unit vectors in 3D

Solution 6

Unit vectors point along the positive coordinate axes. Each has magnitude 1.



In 2D:

$$\mathbf{i} = \begin{pmatrix}1\\0\end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix}0\\1\end{pmatrix}$$

In 3D:

$$\mathbf{i} = \begin{pmatrix}1\\0\\0\end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix}0\\1\\0\end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

Every three-dimensional vector can be written as

$$\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}x\\0\\0\end{pmatrix} + \begin{pmatrix}0\\y\\0\end{pmatrix} + \begin{pmatrix}0\\0\\z\end{pmatrix}$$
$$= x\begin{pmatrix}1\\0\\0\end{pmatrix} + y\begin{pmatrix}0\\1\\0\end{pmatrix} + z\begin{pmatrix}0\\0\\1\end{pmatrix}$$
$$= x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Examples

Solution 7

Recall the 2D unit vectors:

$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$

$8\begin{pmatrix} 1 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \end{pmatrix}$

$= 8\mathbf{i} - 2\mathbf{j}.$

1. Write in $\mathbf{i}, \mathbf{j}, \mathbf{k}$ notation.

a) $(1, 2, 3)^T$

1a $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$

b) $(-7, 3, 0)^T$

1b
$-7\mathbf{i} + 3\mathbf{j}$

c) $(3, 0, -1)^T$

1c
$3\mathbf{i} - \mathbf{k}$

2. Write as a column vector.

a) $4\mathbf{j} + \mathbf{k}$

2a
$\begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix}$

b) $\mathbf{i} - \mathbf{j}$

2b
$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

3. If $A = (1, 2, 3)$, $B = (4, 0, -1)$, find $\overrightarrow{AB}$.

3
$\begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix}$

4. If $\mathbf{a} = (2, 3, 4)^T$, $\mathbf{b} = (0, -1, 3)^T$, find $3\mathbf{a} + 2\mathbf{b}$.

4
$\begin{pmatrix} 6 \\ 7 \\ 18 \end{pmatrix}$

Examples

Solution 8

1. Find the magnitude of $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$, then the unit vector $\hat{\mathbf{a}}$ in its direction.

1
$|\mathbf{a}| = \sqrt{4 + 1 + 16} = \sqrt{21}$

$\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{|\mathbf{a}|} = \dfrac{2\mathbf{i} - \mathbf{j} + 4\mathbf{k}}{\sqrt{21}}$

Dividing by the magnitude gives a vector of unit length without changing its direction.

2. If $\mathbf{a} = (2, -3, 5)^T$ and $\mathbf{b} = (4, -2, 0)^T$, is $2\mathbf{a} - 3\mathbf{b}$ parallel to $4\mathbf{i} - 5\mathbf{k}$?

2
$2\mathbf{a} - 3\mathbf{b} = \begin{pmatrix} -8 \\ 0 \\ 10 \end{pmatrix}$

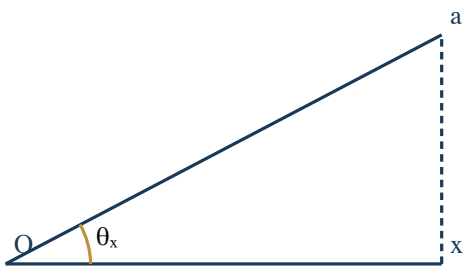
$= -2(4\mathbf{i} - 5\mathbf{k})$

Yes: a non-zero scalar multiple. It points in the opposite direction.

Angles between vectors and an axis

Solution 9

For a non-zero vector $\mathbf{a} = (x, y, z)^T$, use a right-angled triangle to find the angle with a positive coordinate axis.



$$\cos\theta_x = \frac{x}{|\mathbf{a}|}, \quad \cos\theta_y = \frac{y}{|\mathbf{a}|}$$
$$\cos\theta_z = \frac{z}{|\mathbf{a}|}$$

Find the angles that $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ makes with the positive coordinate axes.

Example

$$|\mathbf{a}| = \sqrt{4+9+1} = \sqrt{14}$$
$$\theta_x = \cos^{-1}(2/\sqrt{14}) = 57.7^\circ$$
$$\theta_y = \cos^{-1}(-3/\sqrt{14}) = 143.3^\circ$$
$$\theta_z = \cos^{-1}(-1/\sqrt{14}) = 105.5^\circ$$

Test Your Understanding

Solution 10

1. A and B have position vectors $4\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$ and $3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$. Find $\overrightarrow{AB}$ and show that triangle OAB is isosceles.

1

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (-1, 2, -8)^T$.

$AB = \sqrt{(-1)^2 + 2^2 + (-8)^2}$

$= \sqrt{69}$

$OA = \sqrt{4^2 + 2^2 + 7^2} = \sqrt{69}$

$OB = \sqrt{3^2 + 4^2 + (-1)^2} = \sqrt{26}$.

Since $AB = OA \neq OB$, the triangle is isosceles.

Subtract coordinates directly, or use the difference of position vectors.

2. Find the angle between $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and the positive x -axis.

2

$$|\mathbf{a}| = \sqrt{2^2 + 1^2 + 1^2} = \sqrt{6}$$
$$\theta_a = \cos^{-1}(2/\sqrt{6}) = 35.3^\circ$$

3. Repeat for $\mathbf{b} = \mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$.

3

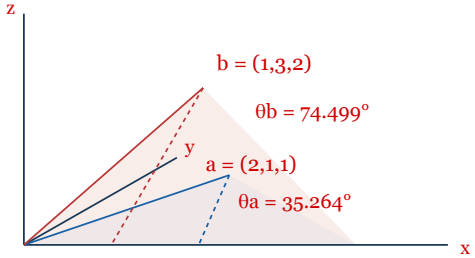
$$|\mathbf{b}| = \sqrt{1^2 + 3^2 + 2^2} = \sqrt{14}$$
$$\theta_b = \cos^{-1}(1/\sqrt{14}) = 74.5^\circ$$

4. Why is the angle between $\mathbf{a}$ and $\mathbf{b}$ not the difference of these angles?

4

The two vectors and the x-axis do not lie in one plane. Their direction angles alone do not determine their mutual angle.

Angle diagram



Two planes through the x-axis (perspective sketch)

13. The position vectors of A, B, C are $(8, -7, 4)^T$, $(8, -3, 3)^T$, $(12, -6, 3)^T$.

a) Find $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{BC}$.

a

$$\overrightarrow{AB} = \begin{pmatrix} 8 \\ -3 \\ 3 \end{pmatrix} - \begin{pmatrix} 8 \\ -7 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ -1 \end{pmatrix}.$$

$$\overrightarrow{AC} = \begin{pmatrix} 12 \\ -6 \\ 3 \end{pmatrix} - \begin{pmatrix} 8 \\ -7 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}.$$

$$\overrightarrow{BC} = \begin{pmatrix} 12 \\ -6 \\ 3 \end{pmatrix} - \begin{pmatrix} 8 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 0 \end{pmatrix}.$$

b) Find the exact lengths of all three sides.

b

$$AB = \sqrt{(0)^2 + (4)^2 + (-1)^2} = \sqrt{17} = \sqrt{17}.$$

$$AC = \sqrt{(4)^2 + (1)^2 + (-1)^2} = \sqrt{18} = 3\sqrt{2}.$$

$$BC = \sqrt{(4)^2 + (-3)^2 + (0)^2} = \sqrt{25} = 5.$$

c) Describe triangle ABC.

c

All three sides have different lengths, so the triangle is scalene.

14. $A = (3, 4, 8)$, $B = (1, -2, 5)$, $C = (7, -5, 7)$.

a) Find $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{BC}$.

a

$$\overrightarrow{AB} = \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} -2 \\ -6 \\ -3 \end{pmatrix}.$$

$$\overrightarrow{AC} = \begin{pmatrix} 7 \\ -5 \\ 7 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} 4 \\ -9 \\ -1 \end{pmatrix}.$$

$$\overrightarrow{BC} = \begin{pmatrix} 7 \\ -5 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 2 \end{pmatrix}.$$

b) Find the side lengths of triangle ABC.

b

$$AB = \sqrt{(-2)^2 + (-6)^2 + (-3)^2} = \sqrt{49} = 7.$$

$$AC = \sqrt{(4)^2 + (-9)^2 + (-1)^2} = \sqrt{98} = 7\sqrt{2}.$$

$$BC = \sqrt{(6)^2 + (-3)^2 + (2)^2} = \sqrt{49} = 7.$$

c) Given $\angle ABC = 90°$, find $\angle BAC$.

c

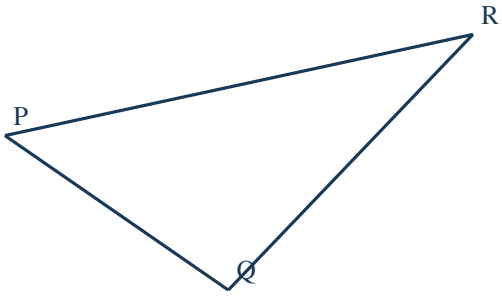
The triangle is right-angled and isosceles.

$\angle BAC = (180° - 90°)/2 = 45°$

Exercise 12B – Y2 Pages 341-343

Solution 13

17. In triangle PQR, $\overrightarrow{PQ} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{QR} = -2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$. Show that $\angle PQR = 78.5°$ to 1 d.p.



Side lengths

$PQ = \sqrt{14}, \quad QR = \sqrt{29}$

$\overrightarrow{PR} = \mathbf{i} + 3\mathbf{j} + 5\mathbf{k}, \quad PR = \sqrt{35}$

Cosine rule

$$\cos \angle PQR = \frac{14 + 29 - 35}{2\sqrt{14}\sqrt{29}}$$

$$= \frac{4}{\sqrt{406}} \Rightarrow \angle PQR = 78.5°$$

Geometric Problems

Solution 14

1. $A = (2, -5, -8)$, $B = (1, -7, -3)$, $C = (0, 15, -10)$, $D = (2, 19, -20)$.

a) Find $\overrightarrow{AB}$ and $\overrightarrow{DC}$ in unit-vector notation.

1a

$\overrightarrow{AB} = -\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$

$\overrightarrow{DC} = -2\mathbf{i} - 4\mathbf{j} + 10\mathbf{k}$

b) Show that $\overrightarrow{DC} = 2\overrightarrow{AB}$, and hence the lines are parallel.

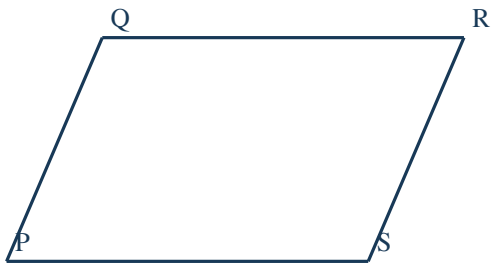
1b $\overrightarrow{DC} = 2(-\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}) = 2\overrightarrow{AB}$

c) Describe quadrilateral $ABCD$.

1c

A trapezium: the opposite sides AB and DC are parallel but unequal.

2. $P = (4, -9, -3)$, $Q = (7, -7, -7)$, $R = (8, -2, 0)$. Find S so that $PQRS$ is a parallelogram.



2

Draw the vertices in sequence. A move from Q to P is the same vector as from R to S.

$\overrightarrow{QP} = (-3, -2, 4)^T$

$\overrightarrow{OS} = \overrightarrow{OR} + \overrightarrow{RS} = \overrightarrow{OR} + \overrightarrow{QP}$

$= (8, -2, 0)^T + (-3, -2, 4)^T = (5, -4, 4)^T$.

Thus $S = (5, -4, 4)$.

Comparing Coefficients

Solution 15

Compare coefficients just as for a polynomial identity: $3x^2 + 5x \equiv A(x^2+1) + Bx + C$ gives $A = 3$ from the x^2 terms.

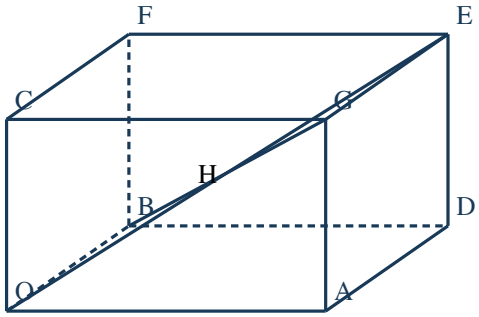
1. Find p, q, r if
$3\mathbf{i} + (p+2)\mathbf{j} + 120\mathbf{k}$
$= p\mathbf{i} - q\mathbf{j} + 4pqr\mathbf{k}$

1

$p = 3, \quad q = -(p+2) = -5$

$120 = 4(3)(-5)r \Rightarrow r = -2$

2. A cuboid has $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$. Prove diagonals OE and BG bisect each other.



2

Find the intersection by two routes and compare coefficients.

$\overrightarrow{OH} = r\overrightarrow{OE} = r(\mathbf{a}+\mathbf{b}+\mathbf{c})$

$= \overrightarrow{OB} + \overrightarrow{BH} = \mathbf{b} + s(\mathbf{a}-\mathbf{b}+\mathbf{c}).$

$r\mathbf{a} + r\mathbf{b} + r\mathbf{c} = s\mathbf{a} + (1-s)\mathbf{b} + s\mathbf{c}.$

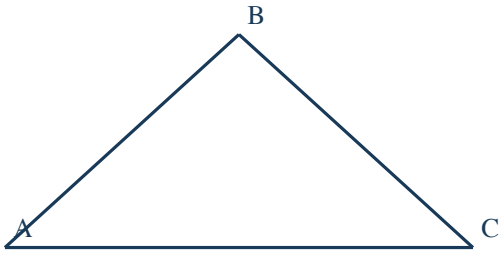
$r = s, \quad r = 1 - s \Rightarrow 2r = 1 \Rightarrow r = s = \frac{1}{2}.$

Thus H is the midpoint of both diagonals.

Exercise 12C – Y2 Pages 346-347

Solution 16

6. Triangle ABC has $\overrightarrow{AB} = 7\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{BC} = -\mathbf{i} + 5\mathbf{k}$.



a) Find its area. (7 marks)

a

$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = (6, -1, 7)^T$

$AB = \sqrt{7^2 + (-1)^2 + 2^2} = \sqrt{54}$

$BC = \sqrt{(-1)^2 + 0^2 + 5^2} = \sqrt{26}$

$AC = \sqrt{6^2 + (-1)^2 + 7^2} = \sqrt{86}$

$\cos B = \dfrac{54 + 26 - 86}{2\sqrt{54}\sqrt{26}} \Rightarrow B = 94.59\ldots^\circ$

$\text{Area} = \frac{1}{2}\sqrt{54}\sqrt{26}\sin B = \dfrac{\sqrt{1395}}{2} \approx 18.67.$

Points D, E satisfy $\overrightarrow{AD} = 3\overrightarrow{AB}$ and $\overrightarrow{AE} = 3\overrightarrow{AC}$.

b) Find the area of triangle ADE. (2 marks)

b

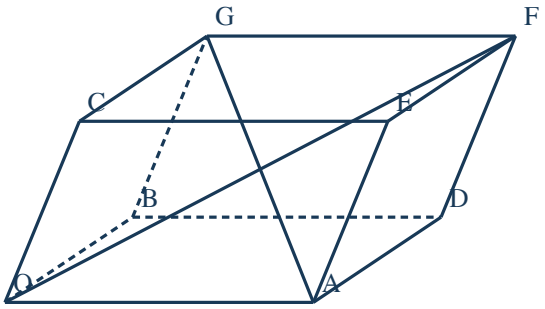
The length scale factor is 3, so the area scale factor is 9.

$\text{Area}(ADE) = \dfrac{9\sqrt{1395}}{2}$

≈ 168.07

Exercise 12C – Y2 Pages 346-347 — Solution 17

7. A parallelepiped is a solid with six parallelogram faces. The diagram has vertices O, A, B, C, D, E, F, G, with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$. Prove that OF and AG bisect each other.



Two routes

Let H be the intersection.

$\overrightarrow{OH} = r\overrightarrow{OF}, \quad \overrightarrow{AH} = s\overrightarrow{AG}$

$\overrightarrow{OH} = \overrightarrow{OA} + \overrightarrow{AH} = \mathbf{a} + s\overrightarrow{AG}.$

$\overrightarrow{OF} = \overrightarrow{OB} + \overrightarrow{BD} + \overrightarrow{DF} = \mathbf{a} + \mathbf{b} + \mathbf{c}$

$\overrightarrow{AG} = \overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{BG} = -\mathbf{a} + \mathbf{b} + \mathbf{c}$

$r(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{a} + s(-\mathbf{a} + \mathbf{b} + \mathbf{c}).$

Conclusion

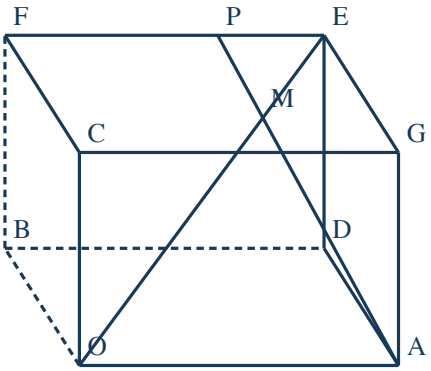
Comparing coefficients of the three independent edge vectors:

$r = 1 - s, \quad r = s \Rightarrow r = s = \frac{1}{2}$

Thus $OH = \frac{1}{2}OF$ and $AH = \frac{1}{2}AG$. The diagonals bisect each other.

Exercise 12C – Y2 Pages 346-347 — Solution 18

8. In the cuboid, $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$. Point M lies on OE, with $OM : ME = 3 : 1$. Line AP passes through M, with $AM : MP = 3 : 1$. Prove that P lies on EF and find $FP : PE$.



Locate P

$\overrightarrow{FP} = \overrightarrow{FB} + \overrightarrow{BO} + \overrightarrow{OA} + \overrightarrow{AP}$

$= -\mathbf{c} - \mathbf{b} + \mathbf{a} + \frac{4}{3}\overrightarrow{AM}.$

$\overrightarrow{AM} = \overrightarrow{AO} + \frac{3}{4}\overrightarrow{OE}$

$= -\mathbf{a} + \frac{3}{4}(\mathbf{a} + \mathbf{b} + \mathbf{c})$

$= -\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b} + \frac{3}{4}\mathbf{c}.$

$\overrightarrow{FP} = -\mathbf{c} - \mathbf{b} + \mathbf{a} + \frac{4}{3}(-\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b} + \frac{3}{4}\mathbf{c}) = \frac{2}{3}\mathbf{a}.$

Position and ratio

$\overrightarrow{PE} = \overrightarrow{PA} + \overrightarrow{AG} + \overrightarrow{GE}$

$= -\frac{4}{3}\overrightarrow{AM} + \mathbf{c} + \mathbf{b}$

$= -\frac{4}{3}(\overrightarrow{AO} + \frac{3}{4}\overrightarrow{OE}) + \mathbf{c} + \mathbf{b}$

$= \frac{4}{3}\mathbf{a} - \mathbf{a} = \frac{1}{3}\mathbf{a}.$

Both vectors are positive multiples of $\overrightarrow{FE} = \mathbf{a}$. Hence P lies on segment FE.

$FP : PE = \frac{2}{3}|\mathbf{a}| : \frac{1}{3}|\mathbf{a}| = 2 : 1.$

Application to Mechanics

Solution 19

Displacement, velocity, acceleration and force are vectors. Speed, mass and time are scalars. Find the magnitudes below.

1. Force: $(3, 4, -1)^T$ N.

[1] $|\mathbf{F}| = \sqrt{26} \approx 5.10\,\text{N}$

2. Acceleration: $(1, 0, 1)^T\,\text{m s}^{-2}$.

[2] $|\mathbf{a}| = \sqrt{2} \approx 1.41\,\text{m s}^{-2}$

3. Displacement: $(12, 3, 4)^T$ m.

[3] $|\mathbf{s}| = \sqrt{144 + 9 + 16} = 13\,\text{m}$

4. Velocity: $(0, 4, 3)^T\,\text{m s}^{-1}$.

[4] $\text{Speed} = \sqrt{16 + 9} = 5\,\text{m s}^{-1}$

The magnitude of displacement is the straight-line separation. It equals distance travelled only for motion along that line without reversing.

Example

Solution 20

A particle of mass 0.5 kg is acted on by three forces:

$\mathbf{F}_1 = (2\mathbf{i} - \mathbf{j} + 2\mathbf{k})\,\text{N}$
$\mathbf{F}_2 = (-\mathbf{i} + 3\mathbf{j} - 3\mathbf{k})\,\text{N}$
$\mathbf{F}_3 = (4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k})\,\text{N}$

a) Find the resultant force.
b) Find the acceleration in unit-vector notation.
c) Find its magnitude.
d) Starting from rest, find the distance travelled in the first 6 seconds.

a
$\mathbf{R} = (2, -1, 2)^T + (-1, 3, -3)^T + (4, -3, -2)^T$
$= (5, -1, -3)^T = (5\mathbf{i} - \mathbf{j} - 3\mathbf{k})\,\text{N}.$

b $\mathbf{a} = \mathbf{R}/0.5 = (10\mathbf{i} - 2\mathbf{j} - 6\mathbf{k})\,\text{m s}^{-2}$

c $|\mathbf{a}| = \sqrt{100 + 4 + 36} = \sqrt{140}\,\text{m s}^{-2}$

d
$s = \tfrac{1}{2}\sqrt{140} \times 6^2 = 36\sqrt{35}\,\text{m}$

Exercise 12D – Y2 Pages 348-349

Solution 21

2. A particle starts from rest and accelerates at $(4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})\,\mathrm{m\,s^{-2}}$ for 2 seconds. Find the distance travelled.

2

$|\mathbf{a}| = \sqrt{16 + 4 + 9} = \sqrt{29}$

$s = \frac{1}{2}\sqrt{29} \times 2^2 = 2\sqrt{29}\,\mathrm{m}$

4. A particle of mass 6 kg is acted on by $\mathbf{F}_1$ and $\mathbf{F}_2$. Given $\mathbf{F}_1 = (7\mathbf{i} + 3\mathbf{j} + \mathbf{k})\,\mathrm{N}$ and $\mathbf{a} = (2\mathbf{i} - \mathbf{k})\,\mathrm{m\,s^{-2}}$, find $\mathbf{F}_2$.

4

$\mathbf{F}_2 = 6(2\mathbf{i} - \mathbf{k}) - (7\mathbf{i} + 3\mathbf{j} + \mathbf{k})$

$= (5\mathbf{i} - 3\mathbf{j} - 7\mathbf{k})\,\mathrm{N}$

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Solution 22

8. The diagram shows parallelogram $ABCD$. Given $\overrightarrow{AB} = (6, -2, 11)^T$ and $\overrightarrow{AC} = (15, 8, 5)^T$, find its area. (7 marks)



Sides

$\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$

$= \begin{pmatrix} 9 \\ 10 \\ -6 \end{pmatrix}$

$AB = \sqrt{161}, \quad AC = \sqrt{314}$

$BC = \sqrt{217}$

Area

$\cos B = \dfrac{161 + 217 - 314}{2\sqrt{161}\sqrt{217}}$

$B \approx 80.1^\circ$

$\text{Area}(ABC) = \frac{1}{2}\sqrt{161}\sqrt{217}\sin B.$

The parallelogram has twice this area:

$\text{Area} = \sqrt{161}\sqrt{217}\sin B$

$= \sqrt{33913} \approx 184$ square units (3 s.f.)